COMP3278

Final Project Tutorial

A walkthrough of the final project workflow

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1 Prepare a work environment (i.e., VS Code)

2 System Design

- Step 1 Information modeling using ER Diagram
- Step 2 Reduce to relational tables
- Step 3 Create the database
- Step 4 Design the SQL to access data for the application

3 Final project requirements

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1 Prepare a work environment (i.e., VS Code)

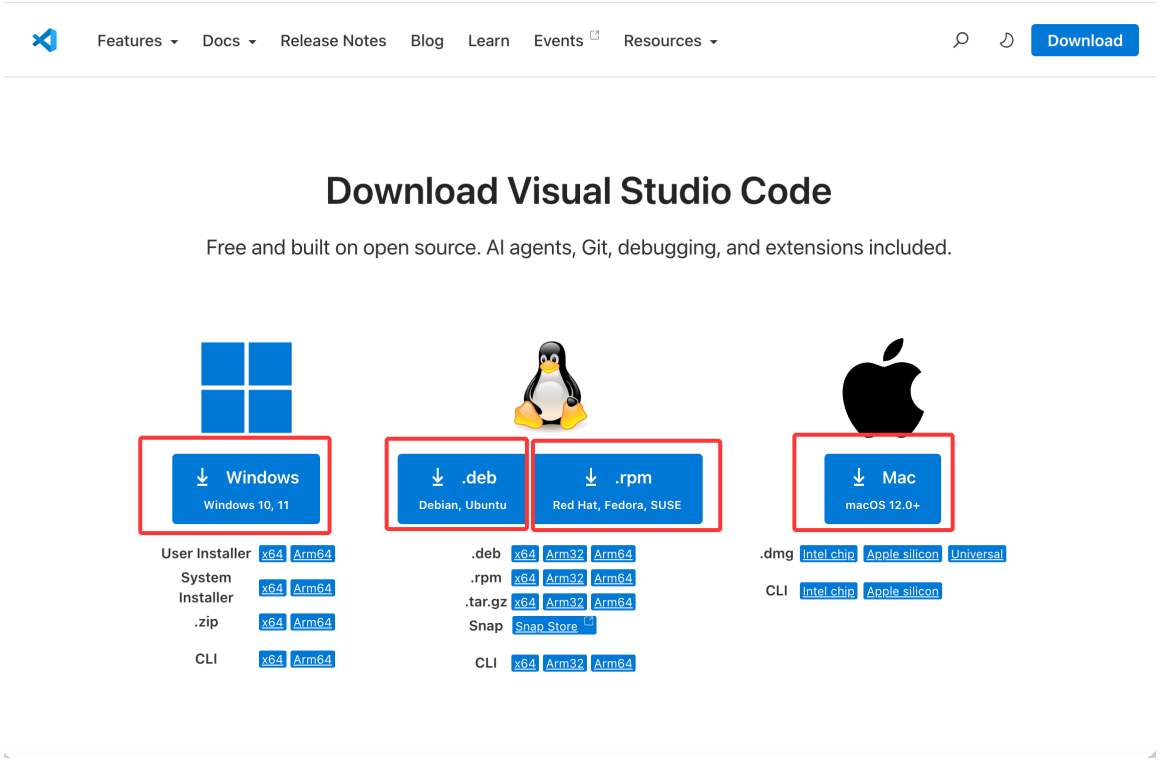
2 System Design

- Step 1 Information modeling using ER Diagram
- Step 2 Reduce to relational tables
- Step 3 Create the database
- Step 4 Design the SQL to access data for the application

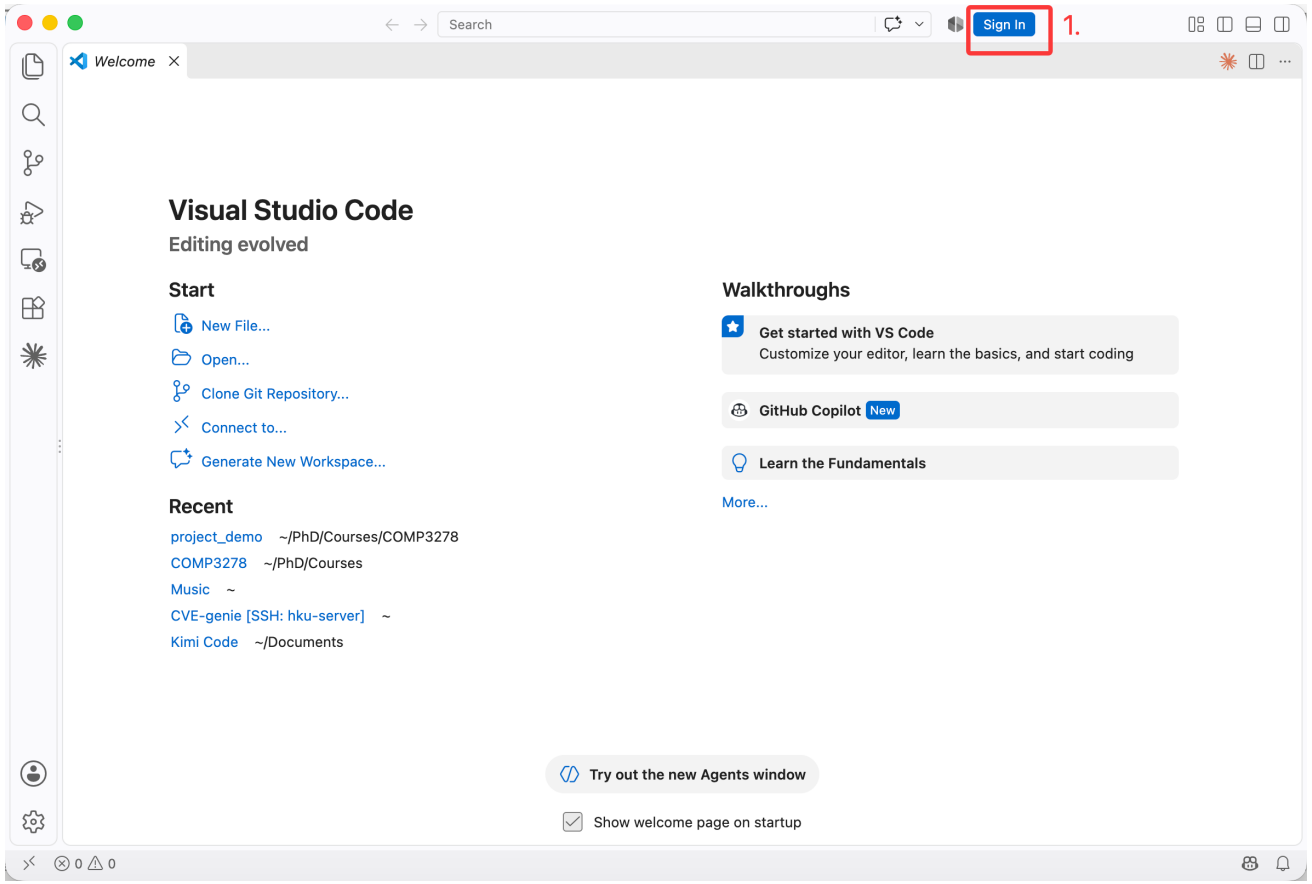
3 Final project requirements

Install Visual Studio Code

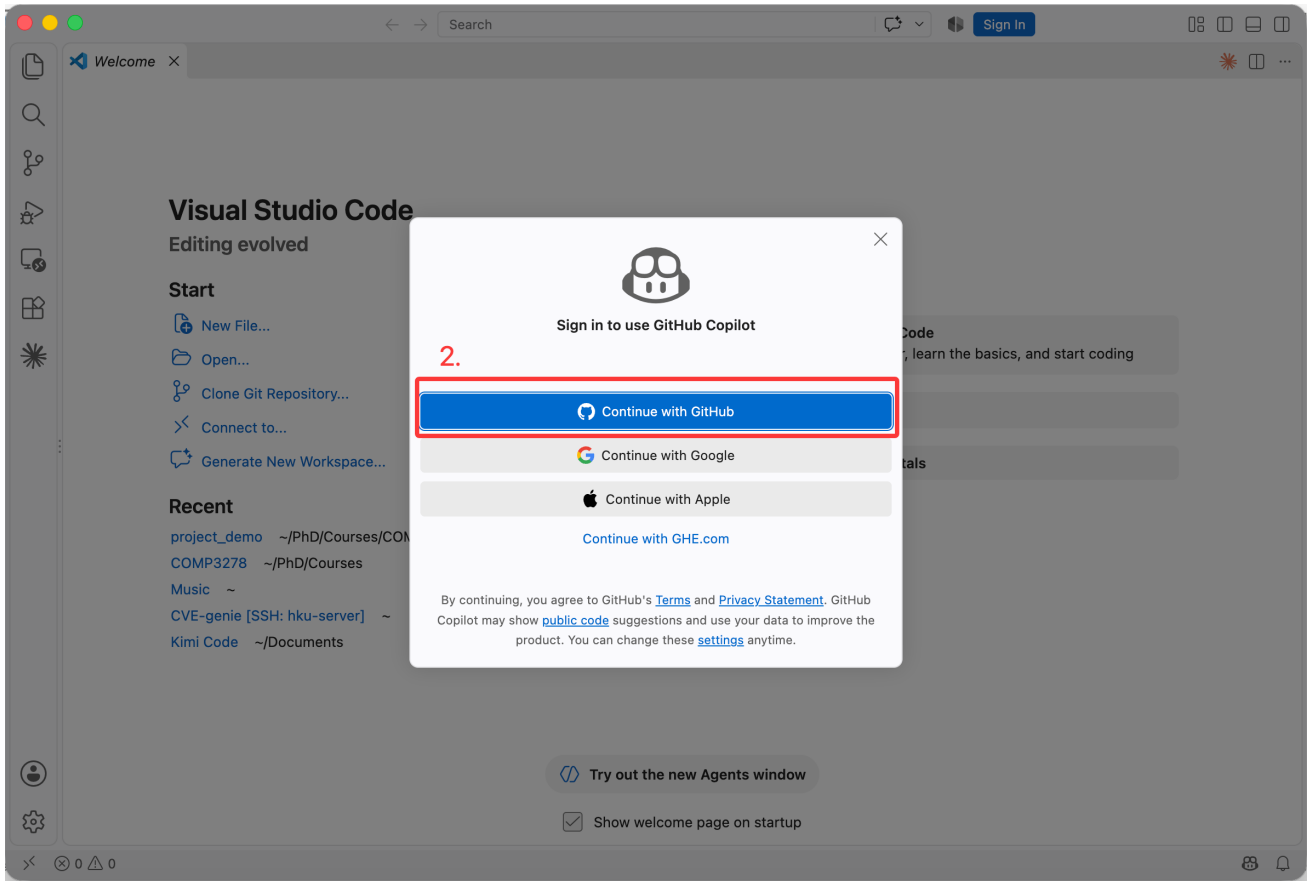
Download the installer for your operating system in the following link:
https://code.visualstudio.com/download



Open VS Code and sign in

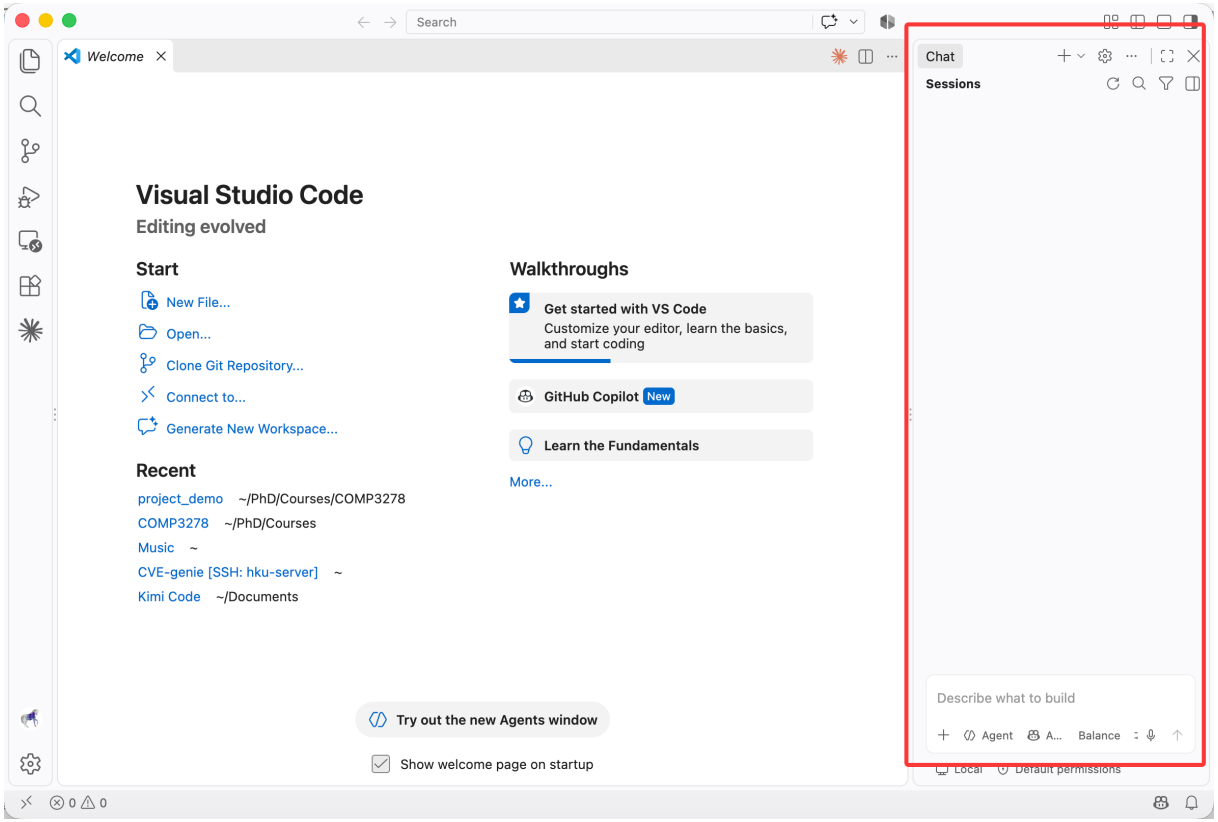


Sign in with GitHub



Copilot will be ready after sign-in

Copilot will occur in right side



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Step 1: Information modeling using ER Diagram

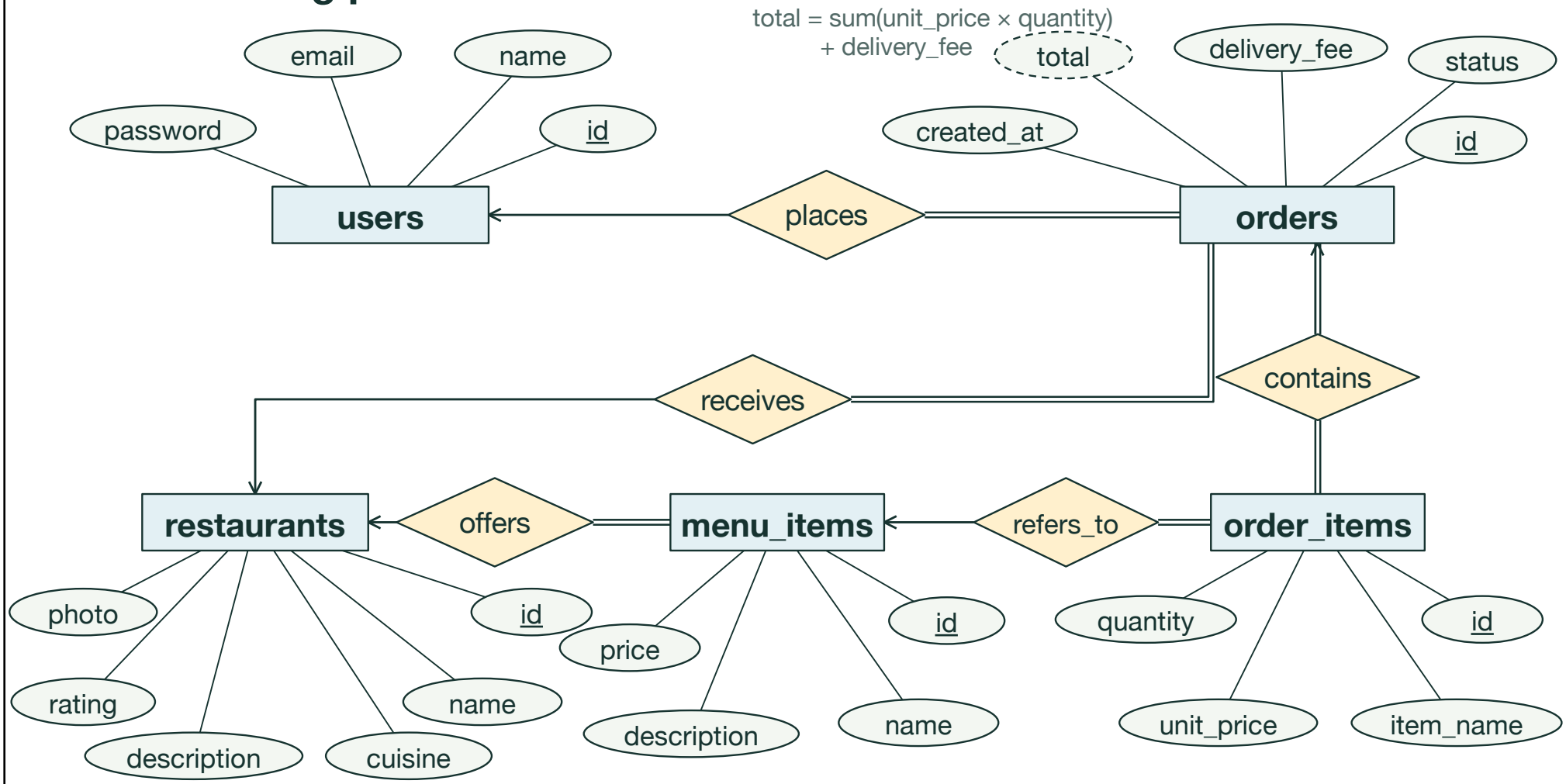
Required functions for a food-ordering platform:

- Register and log in

- Browse restaurants and menus

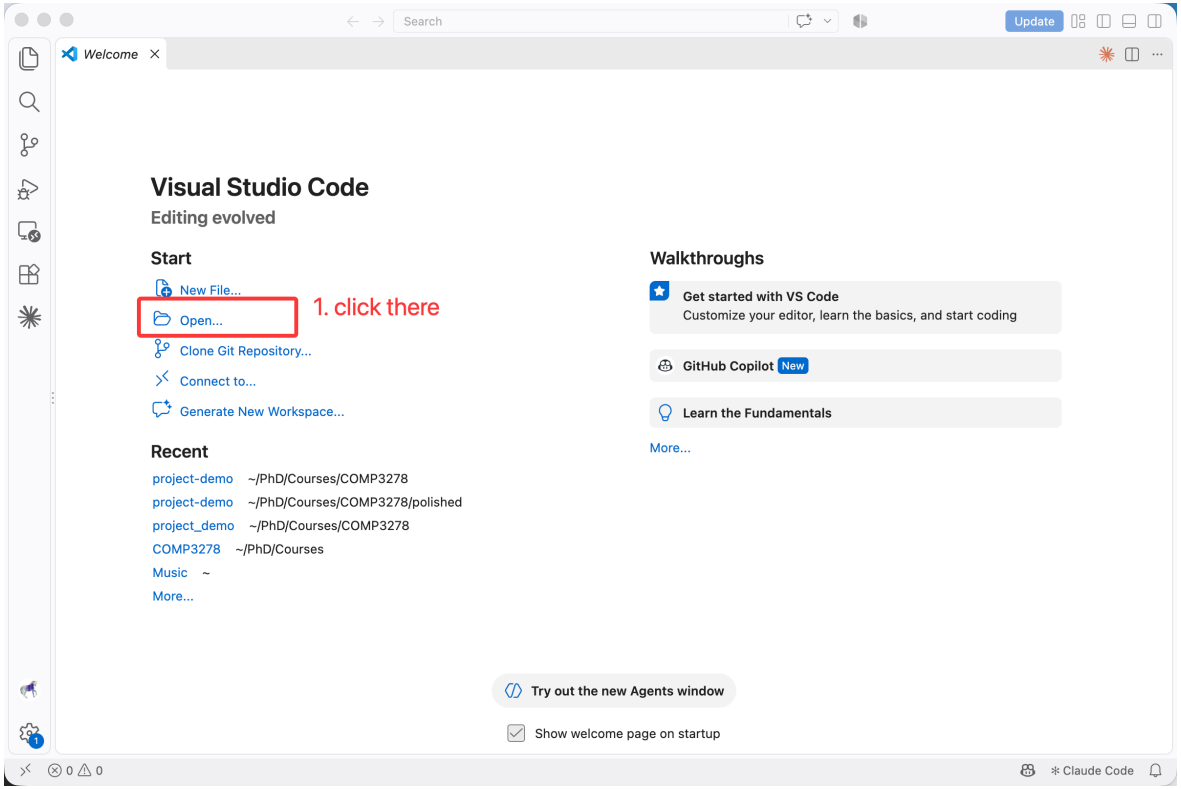
- Users place orders and restaurants receive them

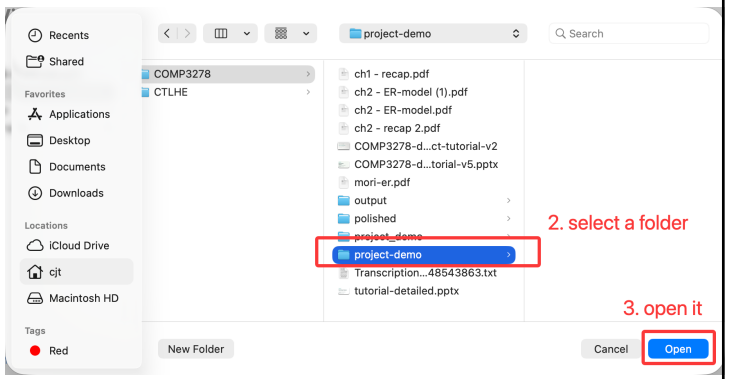
Food-ordering platform: ER model



Step 2: Reduce to relational tables

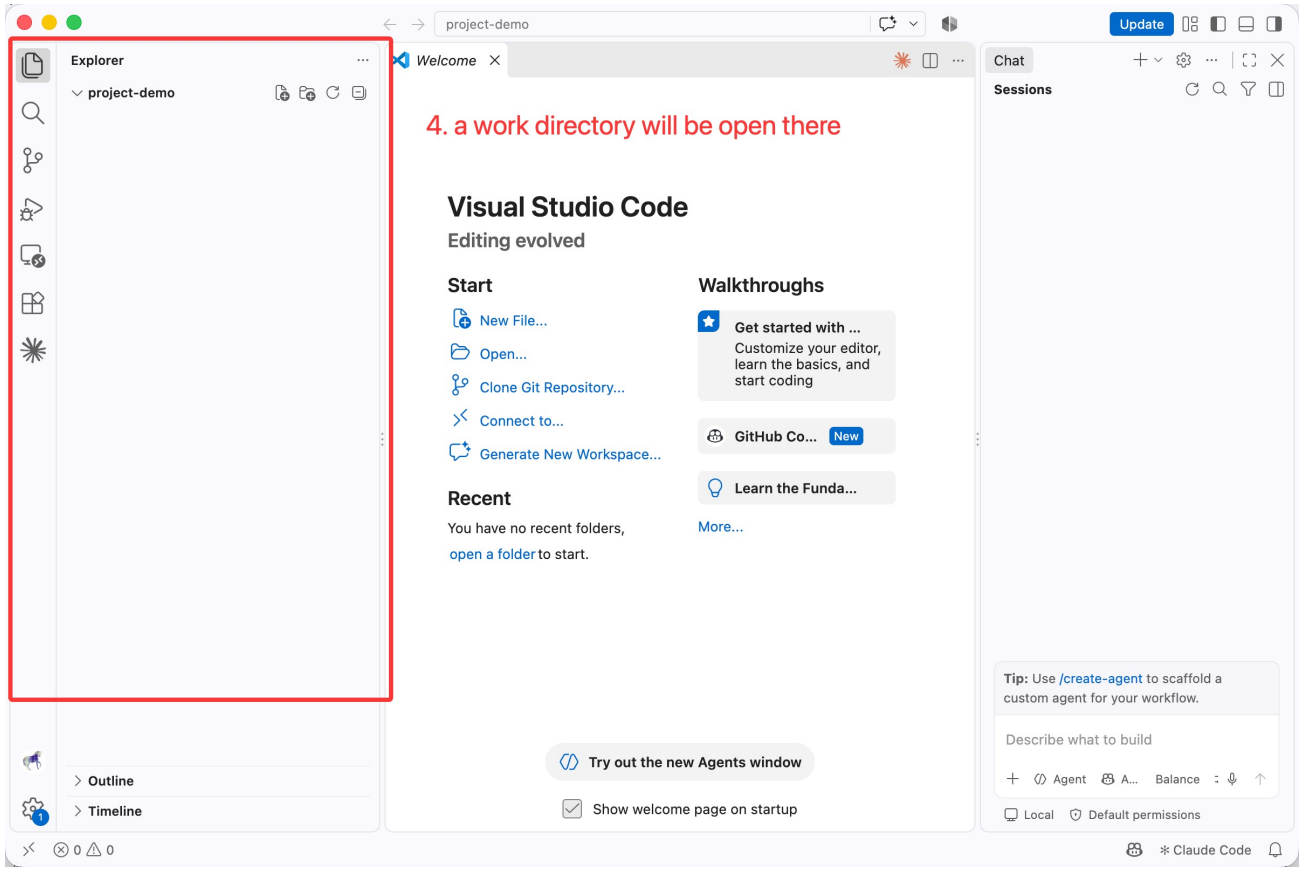
Open a work directory.





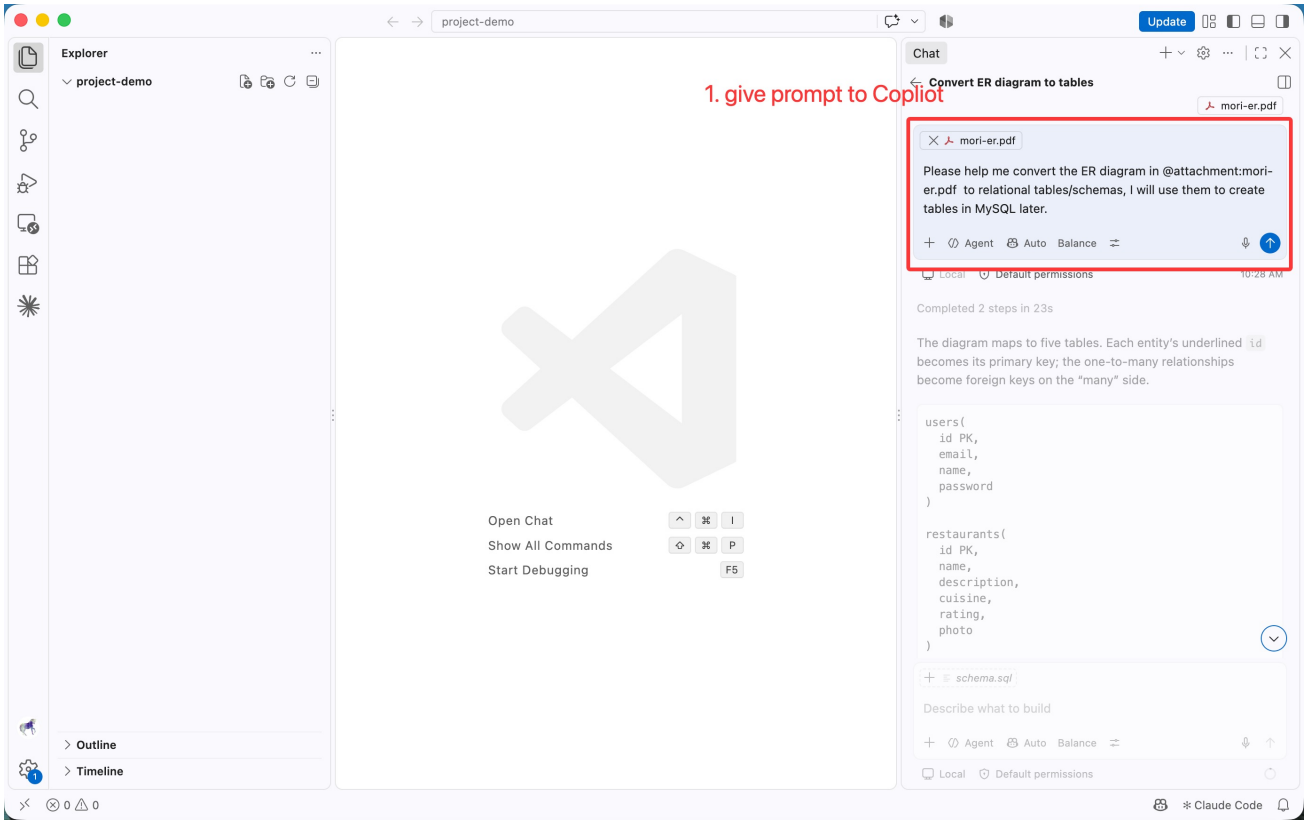
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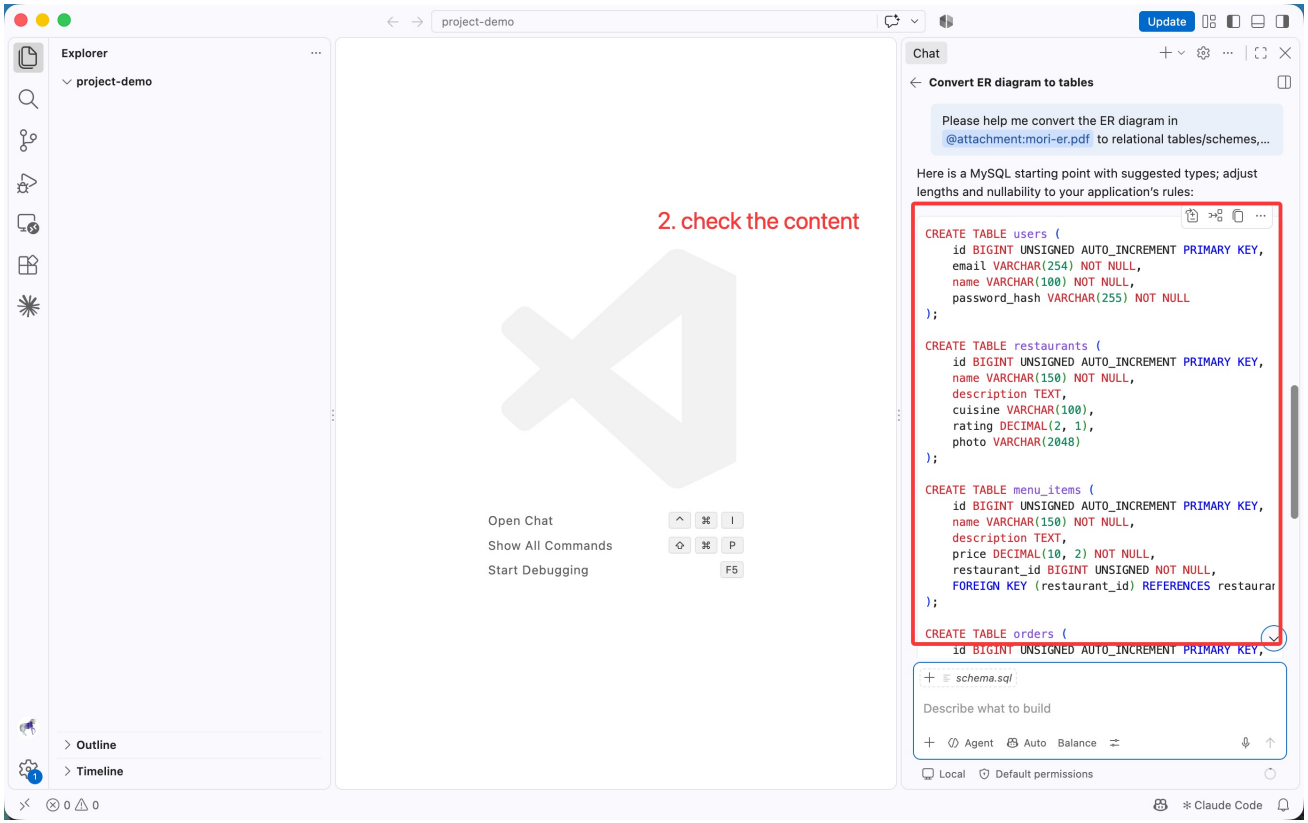
Step 2: Reduce to relational tables

Tell Copilot to reduce the ER diagram to relational tables.



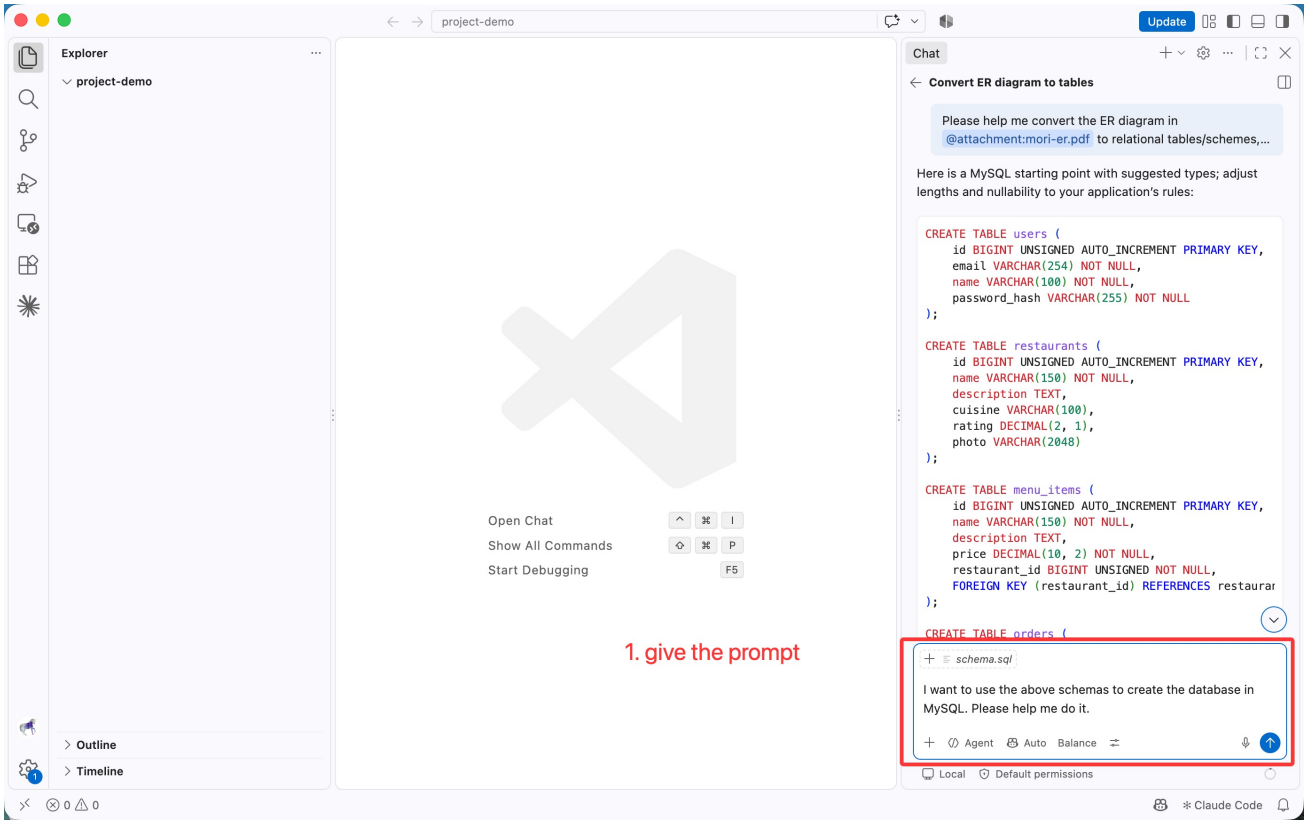
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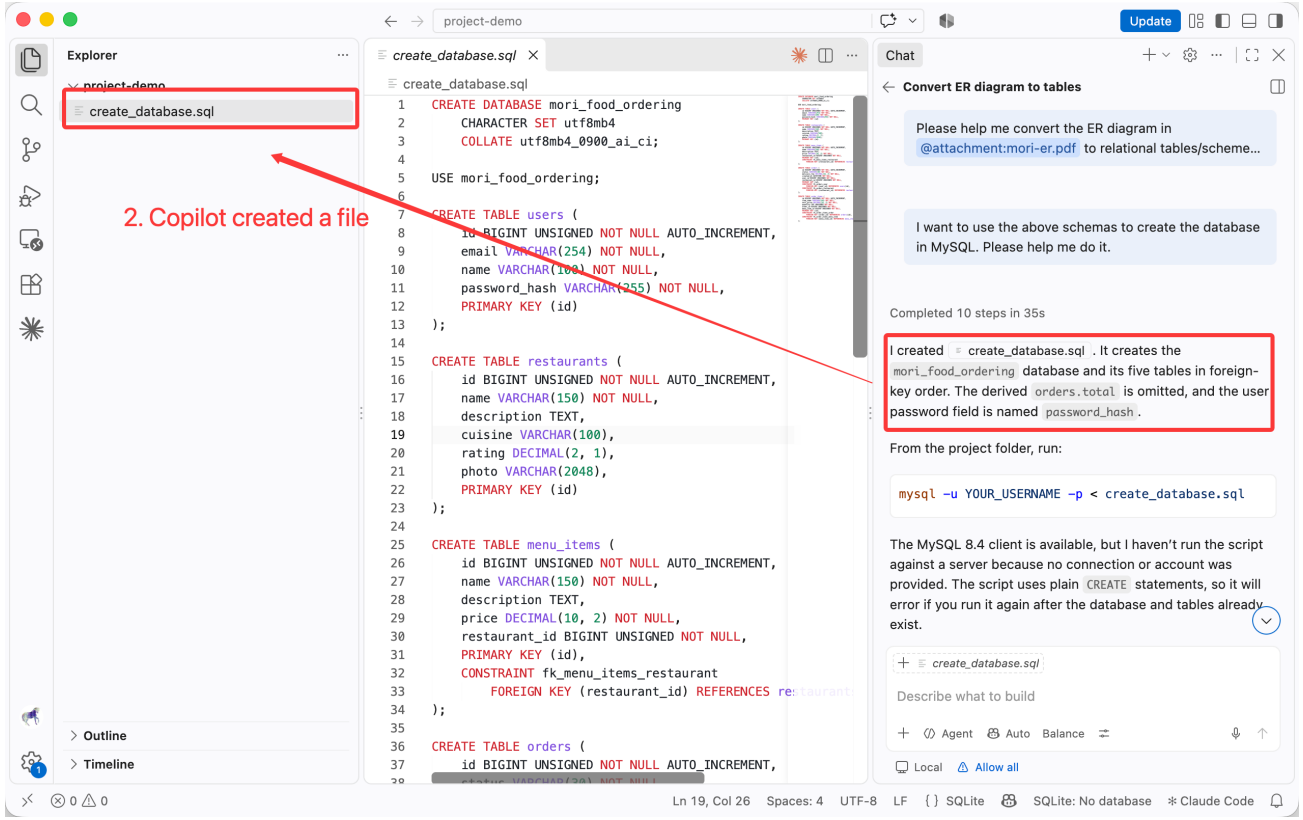
Step 3: Create the database

Ask Copilot to generate the database schema.



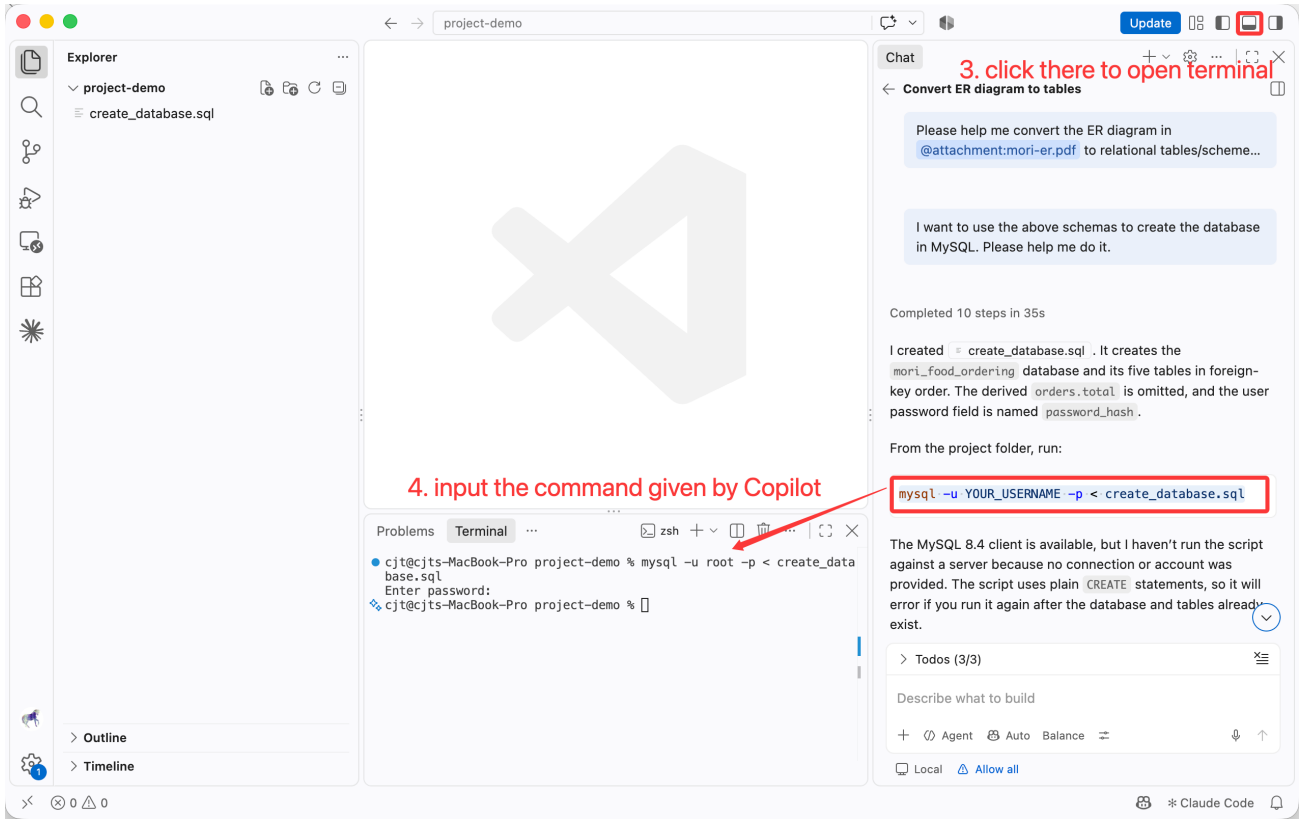
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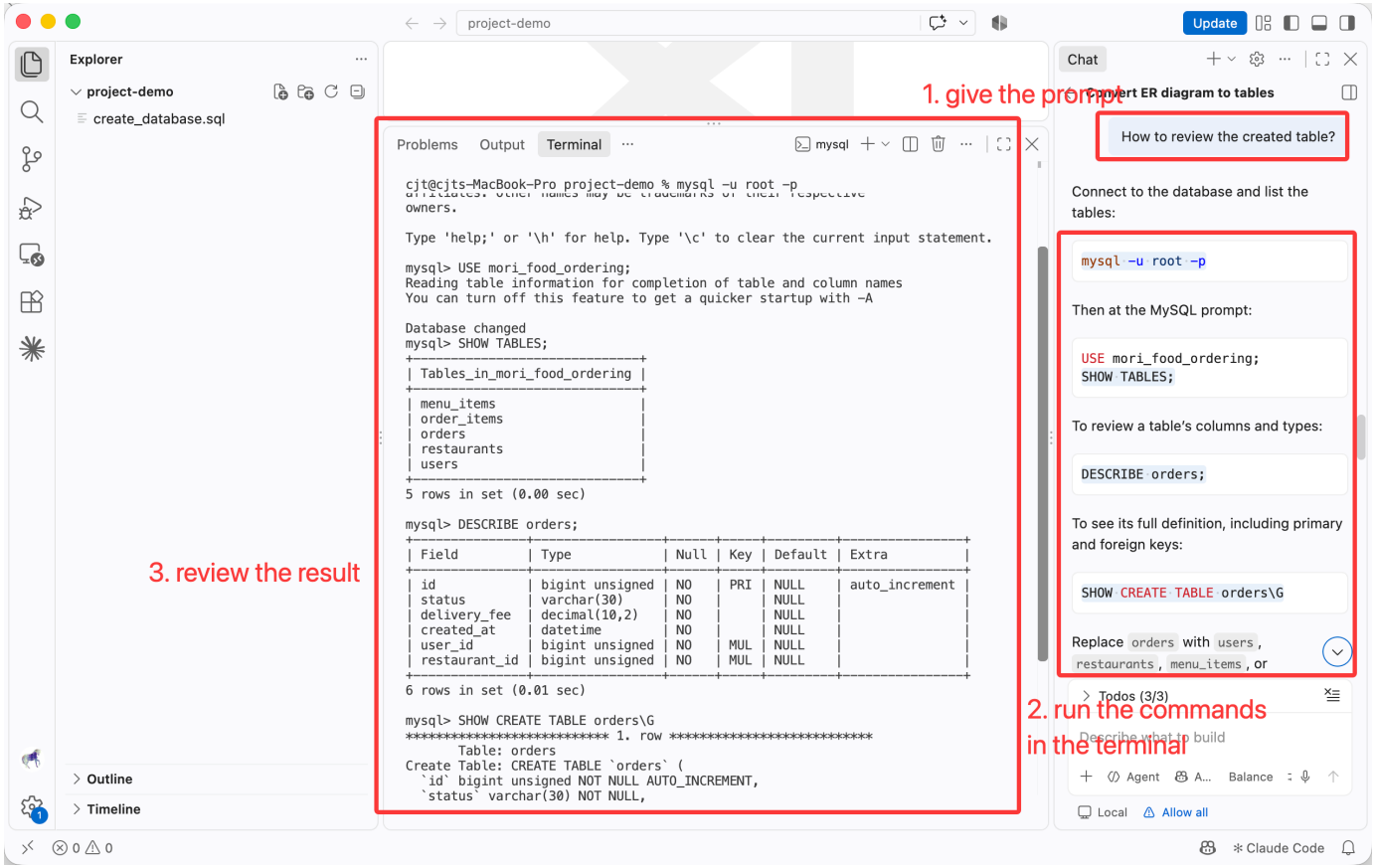
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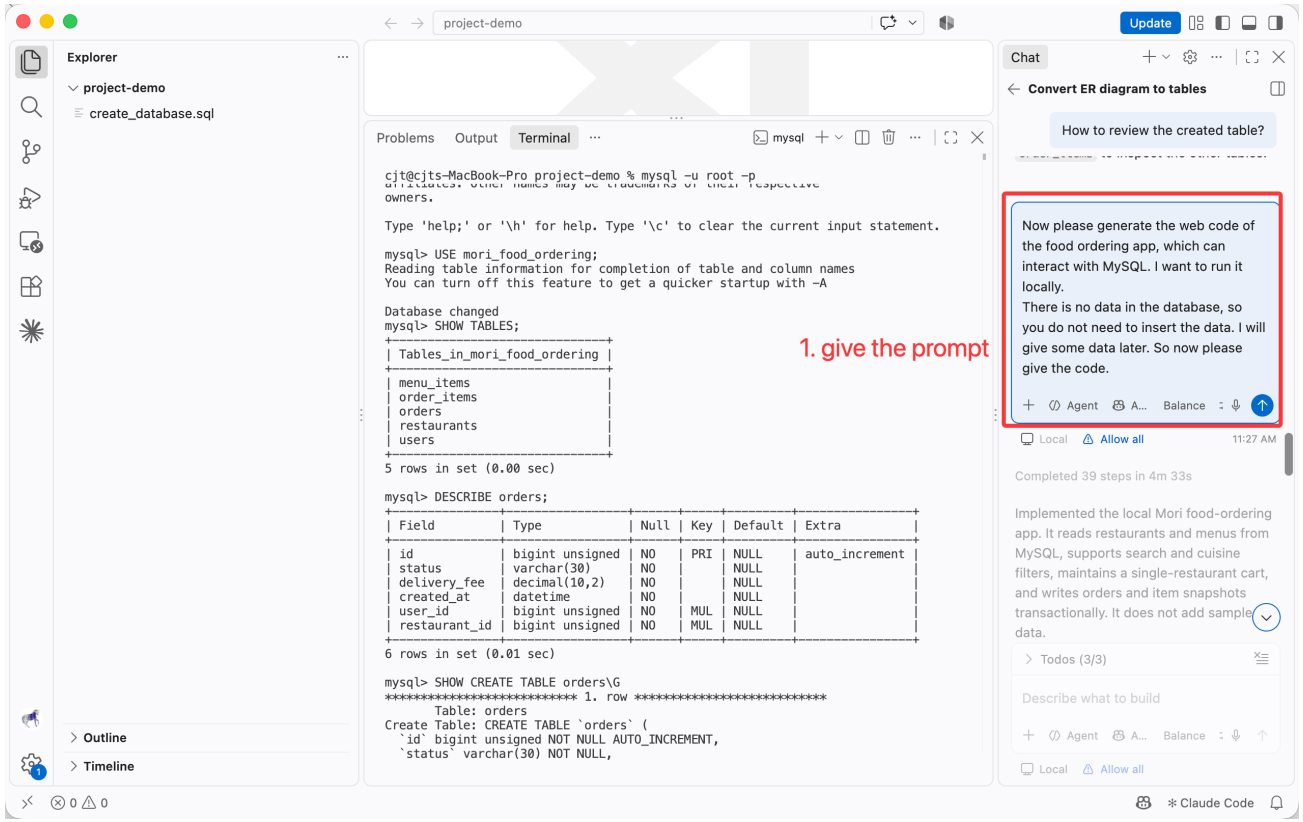
Step 3: Create the database

Manually review the created database.



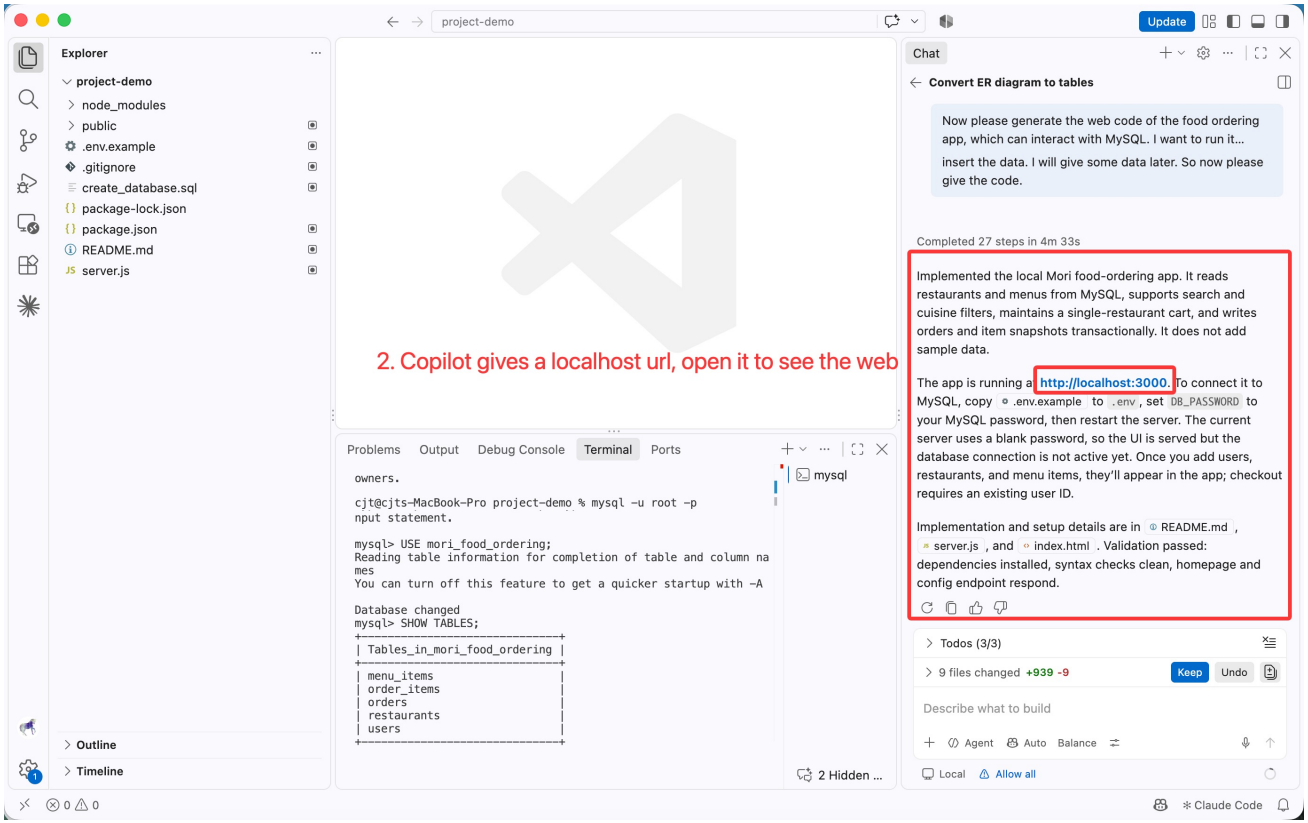
Step 4: Design the SQL to access data for the application

Ask Copilot to generate the web application code.



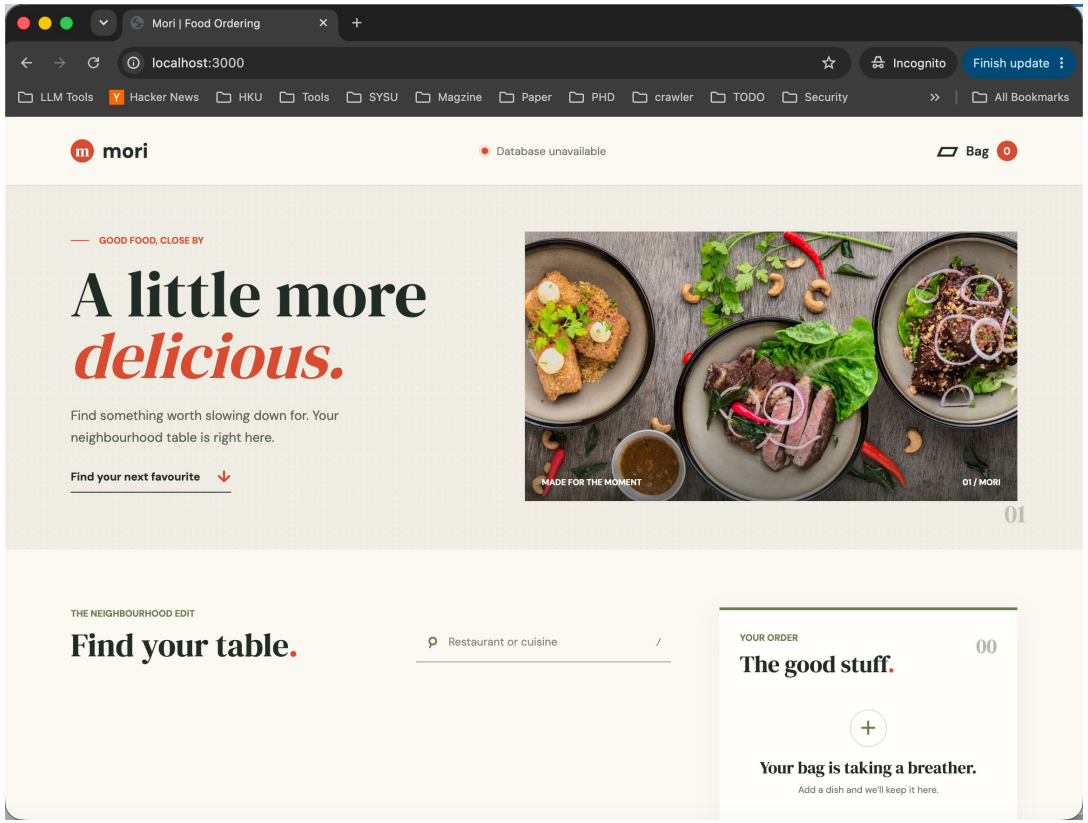
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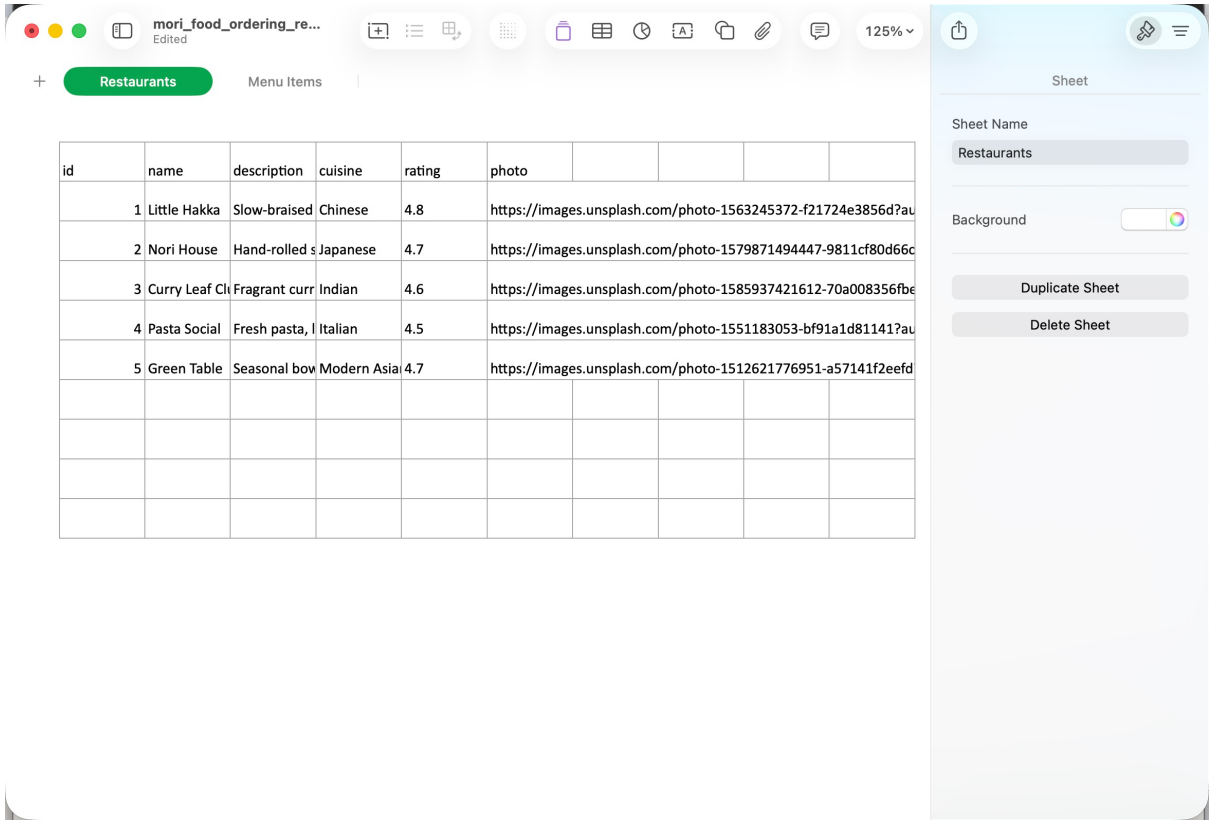
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Opening http://localhost:3000/ in a browser, the app looks good, but there is no data.



Step 4: Design the SQL to access data for the application

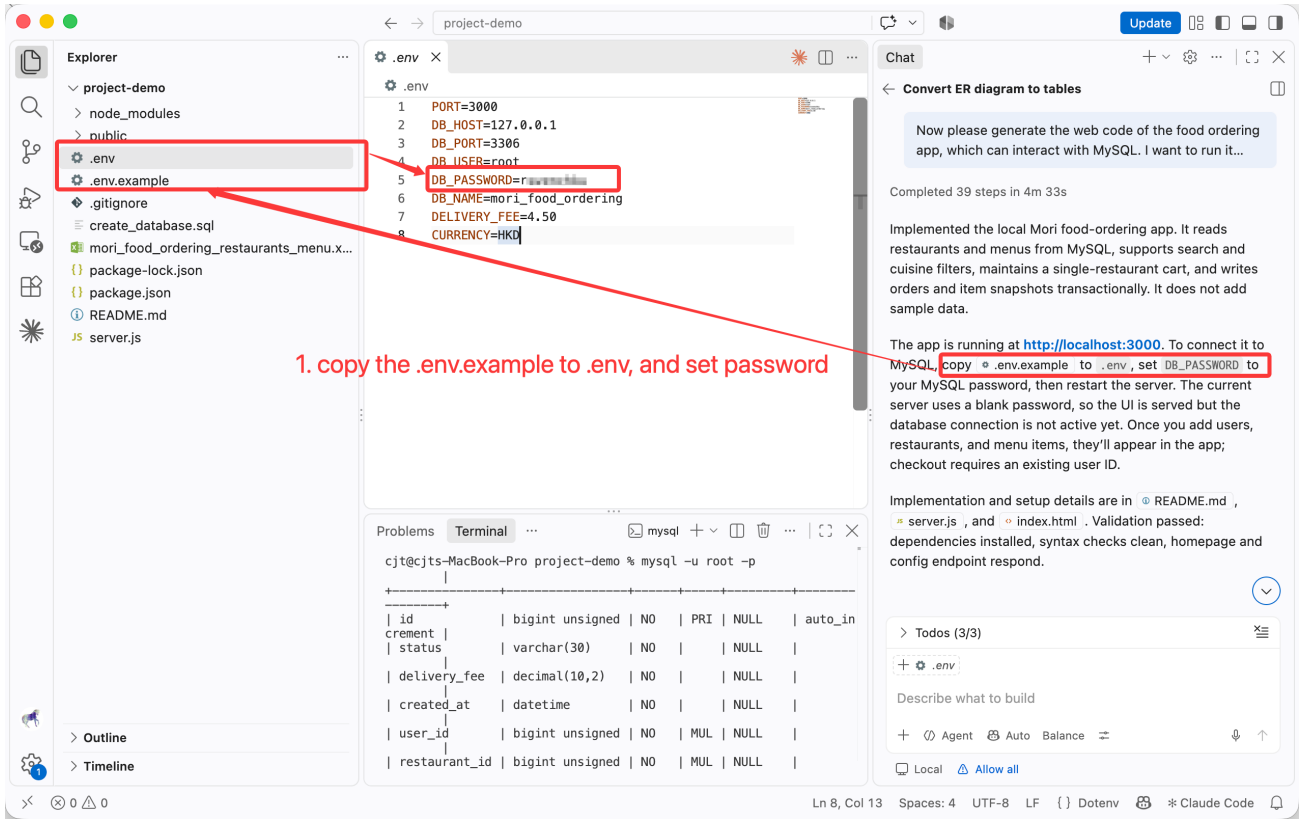
Prepare some data



id	name	description	cuisine	rating	photo
1	Little Hakka	Slow-braised	Chinese	4.8	https://images.unsplash.com/photo-1563245372-f21724e3856d?au
2	Nori House	Hand-rolled s	Japanese	4.7	https://images.unsplash.com/photo-1579871494447-9811cf80d66c
3	Curry Leaf Cl	Fragrant curr	Indian	4.6	https://images.unsplash.com/photo-1585937421612-70a008356fbe
4	Pasta Social	Fresh pasta, l	Italian	4.5	https://images.unsplash.com/photo-1551183053-bf91a1d81141?au
5	Green Table	Seasonal bow	Modern Asia	4.7	https://images.unsplash.com/photo-1512621776951-a57141f2eefd

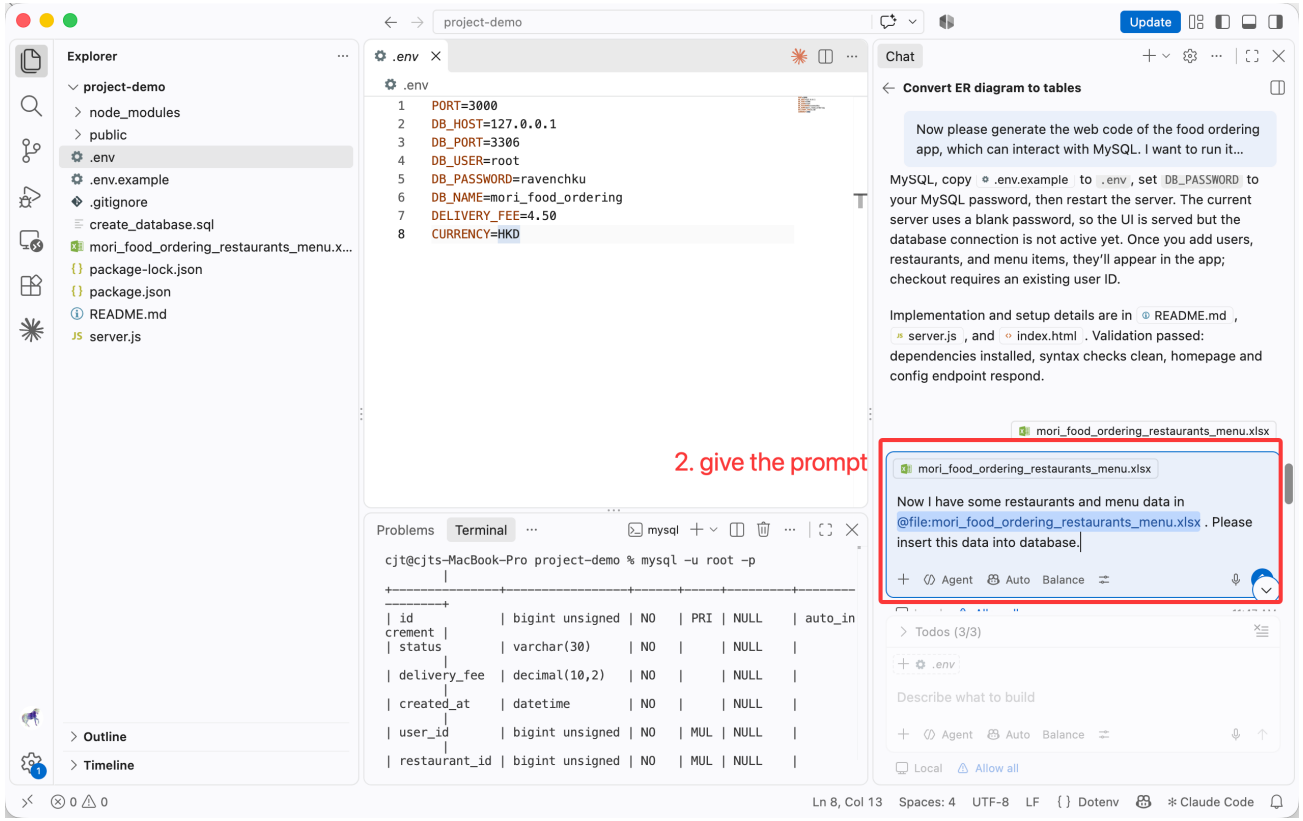
Step 4: Design the SQL to access data for the application

Connect to MySQL.



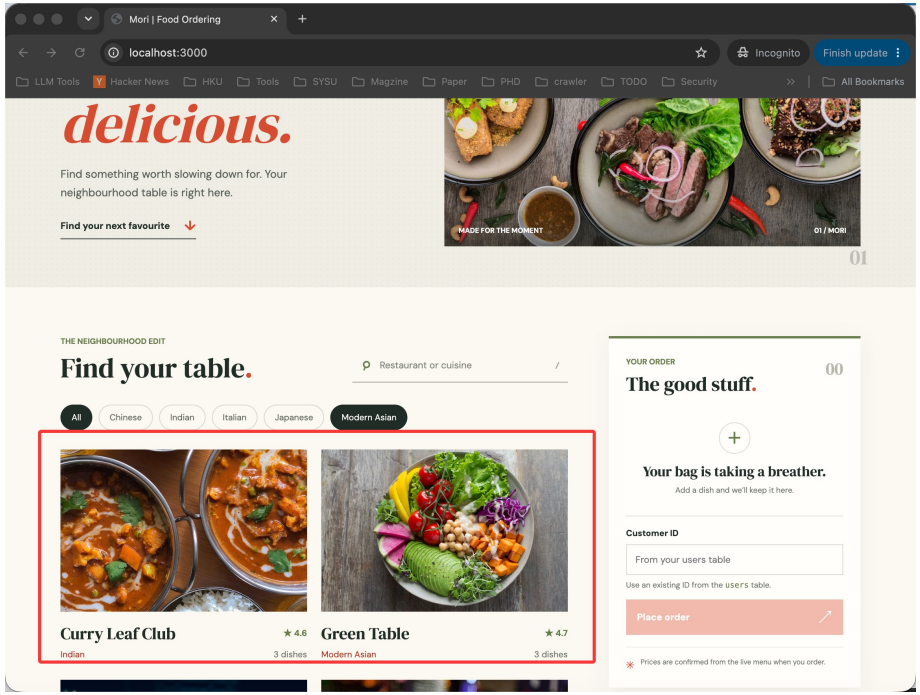
Step 4: Design the SQL to access data for the application

Tell Copilot to insert the data into MySQL.

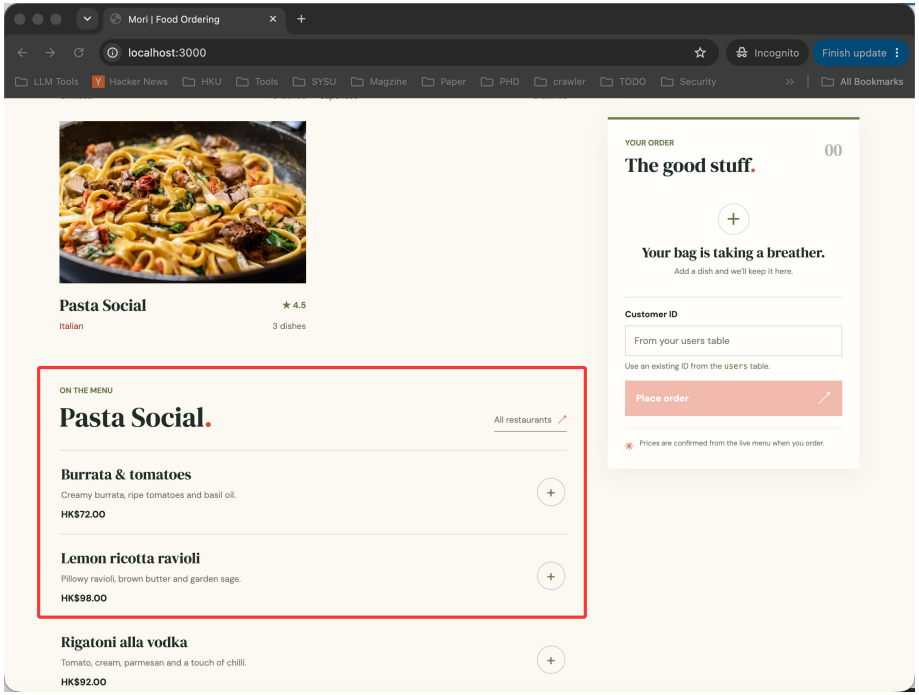


Step 4: Design the SQL to access data for the application

After the inserting data is finished by Copilot, reopen http://localhost:3000/ in browser.



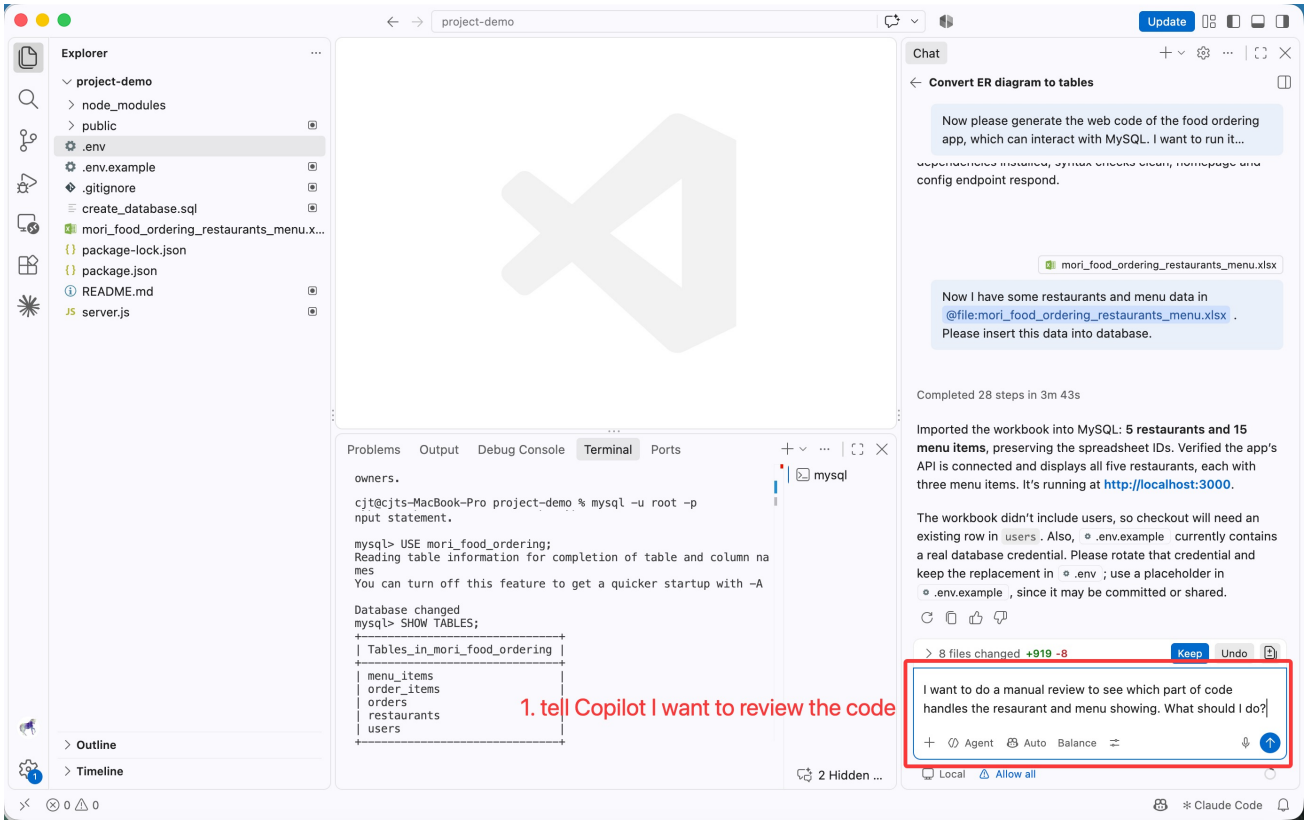
The restaurants appear.



The menus appear.

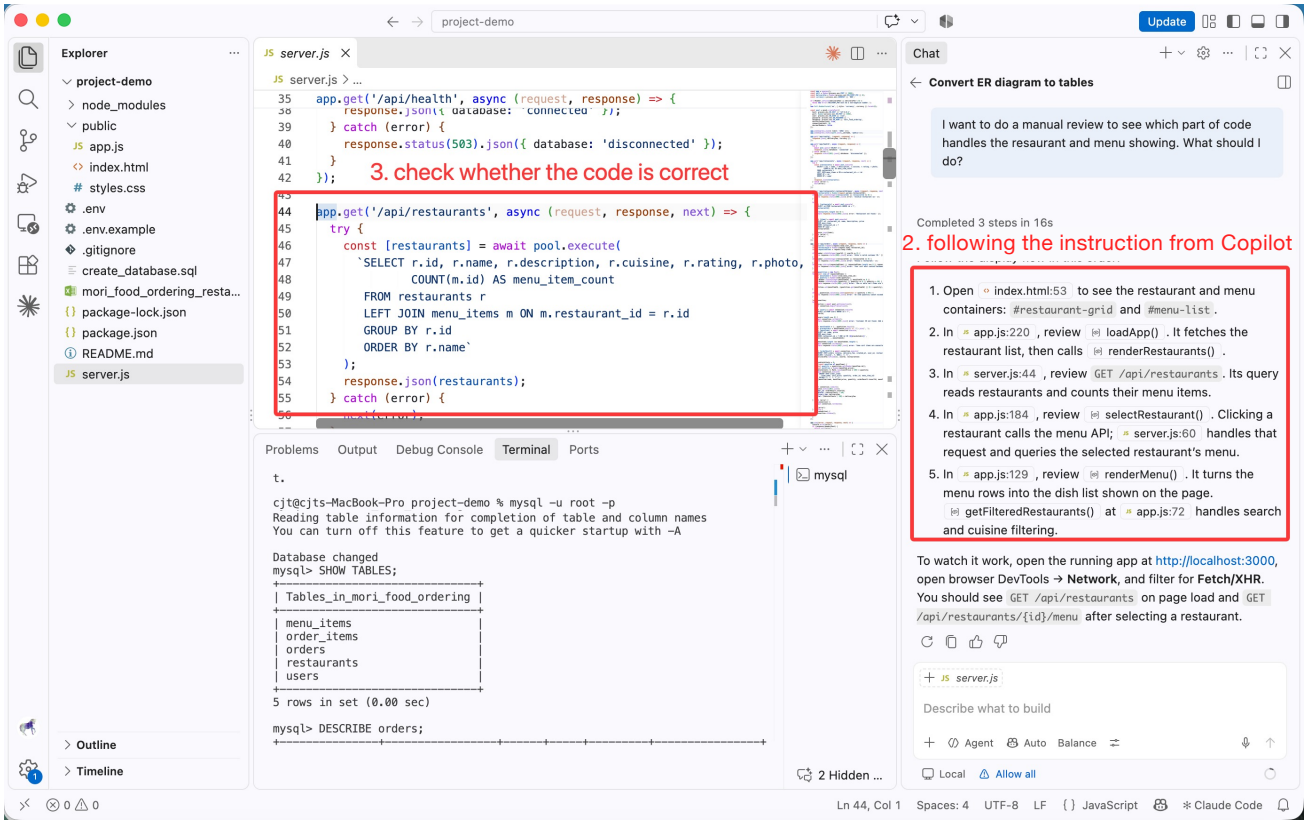
Step 4: Design the SQL to access data for the application

Review the code and the SQL behind each function.



Step 4: Design the SQL to access data for the application

Review the code and the SQL behind each function.



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3 Final project requirements

Final project requirements

Group Information

- Form a group of 3–5 students.
- Email your member list to TA ZHANG Xue (xue.zhang@connect.hku.hk) by **October 15**.
- Students without a group after the deadline will be assigned to groups in student ID order.
- Each group gives a **5-minute** presentation during the final classes.

Project Information

- Choose an app that interests you, such as Steam, Instagram, Amazon, Netflix, Spotify, Uber, Airbnb, Duolingo, Strava, or Keeta, or come up with your own idea. **Implement a few core functions that involve creating, reading, updating, and deleting data in a database (CRUD).**
- Submit your **source code** in Moodle, with a **README** explaining how to run your app and set up its database with data.

Final project requirements

Presentation

- Briefly introduce your web app's functions and design.
- Show the ER diagram.
- Explain the SQL for key functions.
- Include a video demonstrating all or representative functions (maximum 2 minutes). Voice-over explanations are encouraged.
- **Every group member must speak**.